FDDSeg: Unleashing the Power of Scribble Annotation for Cardiac MRI Images Through Feature Decomposition Distillation

Lin Zhang, *Member, IEEE*, Wenzong Li, Kaiyue Bi, Pei Li, Liang Zhang, and Hui Liu, *Member, IEEE*

***Abstract*—Cardiovascular diseases can be diagnosed with computer assistance when using the magnetic resonance imaging (MRI) image that is produced by the MRI sensor. Deep learning-based scribbling MRI image segmentation has demonstrated impressive results recently. However, the majority of current approaches possess an excessive number of model parameters and do not completely utilize scribbling annotations. To develop a feature decomposition distillation deep learning method, named FDDSeg, for scribble-supervised cardiac MRI image segmentation. Public ACDC and MSCMR cardiac MRI datasets were used to evaluate the segmentation performance of FDDSeg. FDDSeg adopts a scribble annotation reuse policy to help provide accurate boundaries, and the intermediate features are split class region and class-free region by using the pseudo labels to further improve feature learning. Effective distillation knowledge is then captured by feature decomposition. FDDSeg was compared with 7 state-of-the-art methods, MAAG, ShapePU, CycleMix, DualBranch, ZscribbleSeg, Perturbation Dual-Branch as well as ScribbleVC on both ACDC and MSCMR datasets. FDDSeg is shown to perform the best in DSC(89.05% and 88.75%), JC(80.30% and 79.78%) as well as HD95(5.76% and 4.44%) metrics with only 2.01 M of parameters. FDDSeg methods can segment cardiac MRI images more precise with only scribble annotations at lower computation cost, which may help increase the efficiency of quantitative analysis of cardiac.**

***Index Terms*—Knowledge distillation, medical image segmentation, weakly supervised learning.**

Received 23 December 2023; revised 10 May 2024; accepted 19 May 2024. Date of publication 24 May 2024; date of current version 8 January 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 61971422, and in part by the Xuzhou Science and Technology Innovation Plan - Key Special Project for Social Development under Grant KC22112. *(Corresponding author: Hui liu.)*

Lin Zhang, Wenzong Li, Kaiyue Bi, and Hui Liu are with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China (e-mail: lin.zhang@cumt.edu.cn; tb22060004a41ld@cumt.edu.cn; hui.liu@cumt.edu.cn).

Pei Li is with the School of Information and Control Engineering, China University of Mining and Technology, Xuzhou 221116, China, and also with the Affiliated Hospital of Xuzhou Medical University, Xuzhou 221000, China (e-mail: xyfylipei@126.com).

Liang Zhang is with the Department of Gastrointestinal Surgery, Xuzhou Central Hospital, Xuzhou 221000, China (e-mail: juwimingz@126.com).

Code and models are available at: https://github.com/labiip/FDDSeg.

Digital Object Identifier 10.1109/JBHI.2024.3404884

I. Introduction

IN CARDIAC quantitative analysis, magnetic resonance imaging (MRI), a non-invasive technique with excellent soft tissue resolution, has emerged as the gold standard for assessing heart function parameters and identifying cardiovascular disorders. Assessing heart functional characteristics requires precise acquisition of cardiac structure. In reality, intracardiac structural calibration is typically done manually by physicians, which takes time, requires a lot of effort, and is subject to the subjective opinion of physicians. The development of an automated cardiac segmentation technique is therefore urgent. Because deep learning has a strong learning and adaptability capacity, it has been employed extensively in medical imaging segmentation in recent years. Based on deep learning, cardiac MRI image segmentation has achieved significant advancements [1], [2].

For deep learning-based image segmentation, there are three primary architectures: fully supervised learning (FSL) [3], [4], unsupervised learning [5], and weakly supervised learning (WSL) [6]. Pixel-by-pixel annotation data is required for supervised segmentation, which is labor-intensive and highly dependent on the expertise of specialists. Unsupervised segmentation is less dependent on data annotation, but it is more challenging to achieve satisfactory segmentation – for example, accurately identifying organizational structures in cardiac MRI images. More researchers prefer weakly guided segmentation because it can produce acceptable segmentation performance based on sparse annotation [7], [8], [9].

Sparse annotation usually contains image-level annotation, extreme points, scribbles, bounding boxes, etc. Among them, scribble displays obvious advantages in annotating complex shape areas such as tissues and organs in a convenient and accurate form, making it a better option for annotating cardiac MRI images [8]. Current scribble segmentation methods on cardiac MRI images mainly focus on data perturbation [10], [11] and post-processing optimization [12], [13]. The data perturbation methods use random occlusion strategies for data augmentation. However, the original image structural information is often destroyed during the process, which does not apply to medical image segmentation tasks since sometimes some minor structural differences may indicate pathological changes. The post-processing optimization methods mainly refine the initial segmentation drawn by the backbone network with an optimization algorithm instead of image destruction. It can be flexibly added to the end of the network, but greatly increases in time and parameter load, which makes it difficult to deploy in practice. With the pre-trained model's supervision, the distillation model can effectively reduce the parameter scale with less cost of accuracy.

To balance the accuracy and parameter scale of automated segmentation, we propose a Feature Decomposition Distillation Model for MRI Image Segmentation (FDDSeg). It mainly focuses on more effective distillation learning and efficient utilization of scribbles. Specifically, the hidden layer features are decomposed into target-related regions and target-free regions by pseudo labels. According to the key learning principle of class-free regions, different attention is given to the decomposed regions for learning, maximizing the utilization of distillation knowledge. In addition, the clustering algorithm and scribble label are combined to generate a pseudo label of unlabeled pixels, which are mixed with network output forming a diversity supervision signal. At the same time, the partial cross-entropy loss is introduced for scribble supervision, to realize the full utilization of scribble annotation. In general, our contributions are summed up as follows:

1) A scribble-reusing (SR) strategy is put forward to obtain sufficient supervisory information from the scribble. The pseudo label, generated by the combination of scribbles and clustering algorithm, is adopted to enrich the boundary-supervised information. By partial of cross entropy loss, the model is directly restricted by scribble annotation to achieve correct finite pixel supervision, thereby locating the main position of the area to be segmented.
2) A Pseudo Label Mixed Supervision (PLMS) module is introduced to effectively integrate the available supervisory information and the constraints learned from the regional level. The pseudo label in 1) and the predictions from both the student branch and teacher branch are dynamically mixed through random weights, supervision information from different perspectives is integrated to provide diverse guidance for the model.
3) A feature decomposition distillation (FDD) module is proposed to refine distillation knowledge and improve distillation efficiency. The features of hidden layers are decomposed into class region and class-free region. Analysis shows that class region is related to shape constraint, while class-free region affects the capture of inter-class relationship. We have revealed the greater importance of inter-class relationship by adjusting the learning weights of two region parts. Then, the class-free region is emphasized to improve feature learning.
4) The performance of FDDSeg is validated on two public cardiac MRI datasets, ACDC and MSCMR. Experimental results show that it outperforms 7 state-of-the-art methods, with a relatively smaller parameter scale.

The framework of this article is arranged as follows. Section II summarizes the related work. The FDDSeg is described in Section III. The dataset, implementation details, evaluation metrics, and experiment results are introduced in Section IV. Further analysis of the experimental results is presented in Section V. Finally, Section VI concludes the article.

II. Related Work

A. Medical Image Segmentation

Medical image segmentation is an important research in clinical medicine. The medical images are usually of uneven pixel intensity and complex gradient changes, resulting in the insufficient recognition of low-level features in images, thus traditional machine learning methods often have limited segmentation performance. The recently emerging deep learning methods have shown excellent performance in image processing tasks with their powerful feature extraction and data fitting capabilities, providing new solutions for medical image segmentation problems. For medical image segmentation, the methods based on Unet and Transformer have performed attractive performance. The Unet [14], an encoder-decoder-based structure with skip connections, is capable of handling medical images with small sample-size well. Its variants such as Unet++[15], Unet3+[1], and Atten-Unet [16] have achieved impressive segmentation through their ability to incorporate either multi-resolution information or attention mechanisms into the feature processing. To model long-range dependencies within data, many scholars try to focus on the Transformer architecture [2], [17]. With self-attention as the key component, the Transformer becomes more capable of dealing with non-local interactions. However, it suffers from large parameters and the Unet-based methods are still widely studied as well as in this paper. Although the mentioned methods have displayed impressive performance, all of them rely on pixel-level annotation which is hard to obtain, so a segmentation method based on limited sparse annotation is needed.

B. Weakly-Supervised Medical Image Segmentation

Recently, the weakly-supervised image segmentation significantly reduces the cost of model training with sparse annotations. Typical annotations contain image-level annotation, extreme points annotation, bounding boxes annotation, and scribble annotation. Based on attention representation learning, Wu et al. [18] developed an image-level weakly-supervised method to segment brain injuries, which improves the performance of segmentation regions. Laradji et al. [19] proposed a weakly-supervised method based on extreme points annotation, which enforces the consistent spatial conversion of the prediction and original image. It could effectively deal with COVID-2019 infected areas segmentation. Zhao et al. [20] developed a bounding boxes annotation segmentation method, which simply needs the 3D bounding box for all instances in the task of 3D instance segmentation. Among the above sparse annotation methods, the scribble annotation is characterized by handling irregular regions and balancing between supervision information and annotation cost, which has greatly attracted the attention of scholars. The current scribble-annotation-based work mainly focuses on improving segmentation performance. For example, Valvano et al. [21] proposed a multi-scale generative adversarial structure, which utilizes a generator to generate segmentation results and mismatched real annotations for identification. This method enhances segmentation performance by fusing multi-scale information, however, suitable real annotations are hard to obtain. Zhang et al. [10], [11] applied a random occlusion strategy to process the original image and forced the model to learn the consistent shape between the processed image and the original image, achieving data augmentation at the cost of losing some structural information in the image, improving the segmentation performance of the model. However, for medical images such as cardiac MRI, the loss of minor structures may result in the missing identification of pathological changes, so the random occlusion strategy is not preferable in our case. Zhang et al. [12] constructed a convolutional neural network to capture spatial prior knowledge by encoding the feature similarities between pixels to adjust the structural shape of the target region. It effectively preserves the structural features of the image, but the computation load is dramatically increased. Luo et al. [14] designed a shared encoder and dual decoder model, which provides additional supervised information by fusing different decoder outputs, thus optimizing the segmentation results. It requires fewer training resources, yet loses target shape constraints. Chen et al. [8] constructed an extra boundary prediction structure to extract the boundaries of the regions to be segmented, with contour information obtained by a pre-trained model offline to constrain its shape, thus achieving accurate boundary segmentation. However, the model parameters are dramatically increased.

C. Knowledge Distillation

The concept of knowledge distillation was initially introduced by Hinton et al. [22], who aimed to accomplish model compression by removing knowledge from the output layer of complicated teacher models and transferring it to simpler student models. Romero et al. [23] improved the performance of the student model by taking into account the limited distillation gain introduced by the output layer and extending knowledge extraction from the output layer to the middle layer for its diversity. Zhao et al. [24] discovered substantial disparities between instance knowledge and background information (DKD). They subsequently transferred knowledge from the pixel level to the instance level, which significantly improved the performance of image classification and object detection. Nevertheless, some researchers have suggested online training self-distillation because the above-mentioned approaches depend on extra pre-training teacher models. To improve the performance of image classification, Zhang et al. [25] used a self-distillation technique in which they transferred knowledge from the output layer to the intermediate layer and deep knowledge to the shallow implementation model. Ji et al. [26] designed an additional top-down network structure to enrich knowledge extraction, providing more refined feature mapping for students while improving the performance of image classification and semantic segmentation. However, these methods can not optimize distillation gain by paying equal attention to different regions of the same feature map.

D. Dual Branch Network

Although the Unet-based methods have achieved some success in medical segmentation, their performance is limited because of the characteristics of medical images including fuzzy structure boundaries and diversity of target scales. Recently, the dual branch network has been proven to be useful to deal with the unstable segmentation caused by medical images themselves. Cheng et al. [27] constructed a dual dense U-shaped network, which employed two densely connected encoders, one of which applied a pre-trained DenseNet as a fixed feature extractor, while the structure of the other was similar to the first one. This method improves segmentation performance by encoding the input image to extract multi-scale semantic information features. Luo et al. [14] proposed a dual decoder structure that can be trained online simultaneously, which better adapts to the distribution of raw data and extracts rich feature information. Inspired by [14], we constructed a dual branch (dual encoder dual decoder) network. We extract more image features through different encoders and assistant train models by dynamically mixing different decoder predictions, which can compensate for the shortcomings of scribble supervision. In addition, based on the dual branch structure, we conveniently implement the FDD strategy that can reduce model parameters while maintaining segmentation performance.

III. Method

With Unet as the backbone [14], FDDSeg adopts the teacher-student distillation structure. It is composed of three modules, including Scribble Reusing (SR), Pseudo Label Mixed Supervision (PLMS), and Feature Decomposition Distillation (FDD), as shown in Fig. 1. The raw image is input into the self-teacher branch and the student branch to extract features $Ft^{(i)}$ and $Fs^{(i)}$ ($i = 1, 2, 3$), along with the predictions $\hat{y_1}$ and $\hat{y_2}$ respectively. On the one hand, scribble information directly supervises $\hat{y_1}$ and $\hat{y_2}$ through partial cross-entropy loss function. On the other hand, it combines the superpixel algorithm to create PL_1. After that, $\hat{y_1}$ and $\hat{y_2}$ are randomly mixed in PLMS to generate PL_2, which is then combined with PL_1 to obtain PL_3. PL_3 is employed to supervise $\hat{y_1}$ and $\hat{y_2}$. Additionally, $Ft^{(i)}$, $Fs^{(i)}$, and PL_2 are input to the FDD module to obtain the decomposition form of $Ft^{(i)}$, $Fs^{(i)}$, including class features and class-free features, as shown in Fig. 2. By minimizing the distance between teacher class features and student class features, as well as between teacher class-free features and student class-free features, we establish teacher guidance for the student. By reusing the scribble annotations, SR provides an accurate boundary for the target area. PLMS integrates pseudo-labels dynamically to obtain dependable monitoring information. Effective distillation knowledge is obtained by FDD through the reinforcement of target-free feature learning because it further captures inter-class correlations.

A. Scribble Reusing

Since scribble annotation can not provide precise shape and boundary information, it is extremely challenging to get ideal supervision if scribble annotation only participates in loss calculation. Thus, we apply the felzenszwalb algorithm to cluster unlabeled pixels into super voxels [28], and assign the super voxel region where scribble annotations are located as pseudo-label $1(PL_1)$. To make full use of the limited known pixel classification in the scribble label, the partial cross-entropy

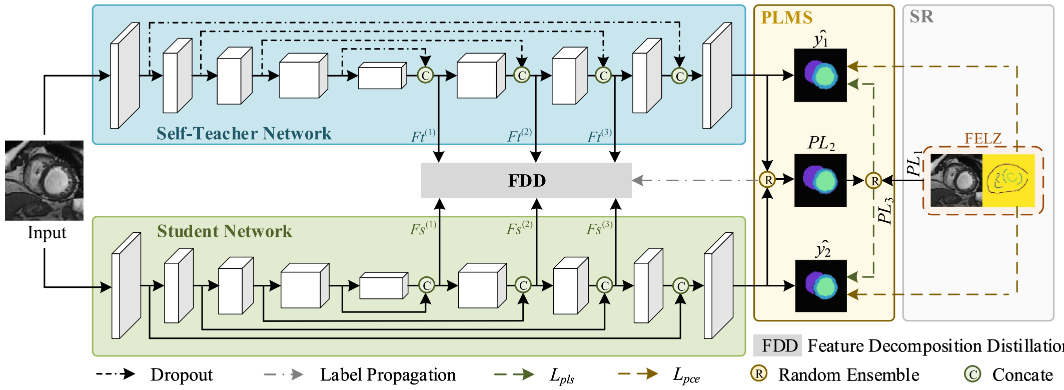


Fig. 1. Overview of proposed FDDSeg. The framework consists of the self-teacher network, the student network, the scribble reusing (SR), the pseudo-label mixed supervision (PLMS), and the feature decomposition distillation (FDD). Notably, the self-teacher and student networks operate with independent weights, with the self-teacher branch incorporating additional dropout layers for feature perturbation. Both the self-teacher prediction $\hat{y_1}$ and the student prediction $\hat{y_2}$ are mixed randomly to be the decomposition label of FDD. Furthermore, the random mixture of PL_2 and PL_1, as well as the scribble annotation is the supervision signal for the self-teacher network and student network.

loss is introduced [14]

$$L_{pce} = L'_{pce}(s, \hat{y}_1) + L'_{pce}(s, \hat{y}_2). \tag{1}$$

where L'_{pce} is the partial cross-entropy loss, s indicates the scribble annotations, $\hat{y}_1$ and $\hat{y}_2$ represent the prediction of the teacher and the student, respectively. On the one hand, SR directly utilizes scribbles as supervision, offering a restricted yet accurate level of supervision. On the other hand, it combines scribble information with superpixel algorithms to generate supervision pseudo labels for regional-level supervision. The pseudo label is independent of model parameter optimization, providing supervisory information from different perspectives.

B. Pseudo Label Mixed Supervision

Pseudo labeling [14], [29] is proved to work for boosting performance in weakly-supervised learning. Thus, the predictions of different branches are dynamically mixed to generate pseudo label PL_2. Since the generation of PL_1 is independent of model training and does not reject model parameter updates, we further mix PL_1 and PL_2 dynamically to obtain PL_3, which is used as final supervision label. Thus, we first fuse the prediction of both the teacher and the student to form the pseudo-label 2 (PL_2)

$$PL_2 = \beta \times \hat{y}_1 + (1 - \beta) \times \hat{y}_2. \tag{2}$$

where $\beta \in (0, 1)$ is updated after every iteration. The pseudo-label 3 (PL_3) is created by the mixing of PL_1 and PL_2

$$PL_3 = \gamma \times PL_1 + (1 - \gamma) \times PL_2. \tag{3}$$

where $\gamma \in (0, 1)$ is updated after every iteration. Applying PL_3 to supervise teacher and student branches

$$L_{pls} = L_{dice}(PL_3, \hat{y}_1) + L_{dice}(PL_3, \hat{y}_2). \tag{4}$$

where L_{dice} is the dice loss. PLMS also contributes to regional-level supervision by dynamically integrating information from three types of pseudo labels, thereby simplifying the complexity of parameter optimization. We consider that the model can assimilate the distinctive features of these labels after adequate iterations.

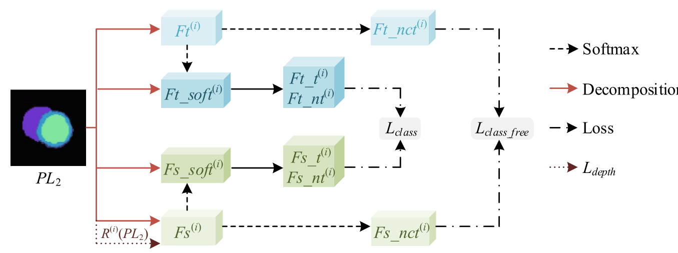


Fig. 2. Featrure decomposition distillation (FDD). The figure illustrates the relationship between the features pre and post-decomposition within the teacher and student branches in FDD, and the constraints between the teacher class feature $\{Ft\_t^{(}i),\ Ft\_nt^{(}i)\}$ and student class feature $\{Fs\_t^{(}i),\ Fs\_nt^{(}i)\}$ as well as the constraints between the teacher class-free feature $Ft\_nct^{(}i)$ and student class-free feature $Fs\_nct^{(}i)$.

C. Feature Decomposition Distillation

Inspired by DKD [24], we found that different feature regions play different roles in the distillation process. To be more specific, assume it's a n-classes classification task (n is 4 in this paper), the teacher and the student exhibit different feature responses at n public-free areas m_1 to m_4 in the class region (student features show high response in the m_1 area and low response in the m_2 area; teacher features show low response in the m_1 area and high response in the m_2 area, etc.). The teacher can have shape constraints on the student through the differences in feature responses in the above four areas. For a certain pixel, the teacher and student generate a 4-class prediction, respectively. In the class-free region, there is an implicit inter-class dependency relationship between these three types of class-free predictions and the corresponding class predictions, and it can also be depicted by the three types of class-free predictions. Therefore, the relationship can be used as knowledge for learning, as shown in Fig. 3.

Based on the above analysis, an additional Feature Decomposition Distillation module (FDD) is designed in the middle feature layer to enhance the extraction of semantic features of cardiac structure from different scales. Firstly, the teacher feature $Ft^{(i)}$ and student feature $Fs^{(i)}$ (i=1,2,3) are decomposed into

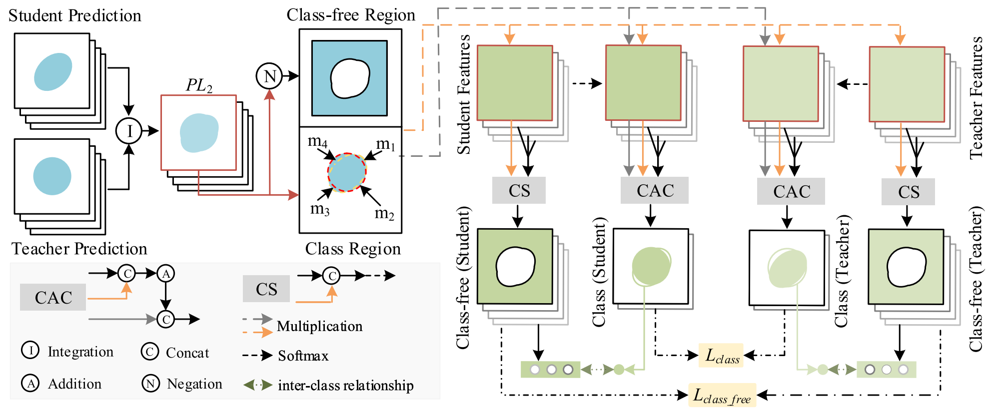


Fig. 3. Specific feature decomposition distillation. Here, we take a class of targets in 4-object segmentation as an example to illustrate the specific decomposition for teacher and student features via PL_2. The decomposition approach for the remaining three types of objectives aligns consistently with the methodology outlined in this figure.

class features ($\{Ft\_t^{(i)}, Ft\_nt^{(i)}\}$, $\{Fs\_t^{(i)}, Fs\_nt^{(i)}\}$) and class-free features ($Ft\_nct^{(i)}$, $Fs\_nct^{(i)}$) by PL_2, respectively. The class features are reflected in binary classification, where $Ft\_t^{(i)}$ represents the current class segmentation foreground and is used for explicit shape constraint. $Ft\_nt^{(i)}$ indicates the current class segmentation background. It contains the fusion of foreground probabilities for the remaining classes. Note that the four types of foreground correspond to four different backgrounds. This transition from foreground to background is beneficial for providing richer feature representation. On the other hand, class-free features are expressed in a multi-classification format, with $Ft\_nct^{(i)}$ representing a redistribution of the predicted probability aimed at excluding the current segmented foreground. The same principle applies to student features.Then these two types of features are distilled separately

$$\begin{aligned}
L_{fdd} &= \sum_{i=1}^{3}((1-\alpha)\times L_{class}^{(i)} + \alpha \times L_{class_free}^{(i)}) \\
&= \alpha \times \sum_{i=1}^{3} L_{mse}(Ft_nct^{(i)}, Fs_nct^{(i)}) \\
&\quad + (1-\alpha)\times \sum_{i=1}^{3} L_{mse}(Ft_t^{(i)}, Fs_t^{(i)}) \\
&\quad + (1-\alpha)\times \sum_{i=1}^{3} L_{mse}(Ft_nt^{(i)}, Fs_nt^{(i)}). \qquad (5)
\end{aligned}$$

where L_{mse} is the mse loss, $L_{class\_free}^{(i)}$ represents the distillation of class-free features with α weight, while $L_{class}^{(i)}$ represents the distillation of class features of $1-\alpha$ weight. The differences between FDD and DKD are as follows: (1) DKD is designed for classification task, while FDD is for weakly supervised segmentation task. (2) To obtain knowledge that is beneficial for segmentation task, FDD acts on the intermediate feature layer instead of the final output layer. (3) FDD applies mse loss rather than kl loss. This choice stems from the structural similarity between the teacher and student branches in our design, yielding predictions that closely resemble each other. Consequently, kl loss struggles to quantify differences between these similar predictions, whereas mse loss effectively minimizes discrepancies at the pixel level, facilitating FDD's iterative learning process. In addition, by scaling PL_2, we generate supervision labels $R^{(i)}(PL_2)$ for each scale, and deep supervision between $R^{(i)}(PL_2)$ and $Fs^{(i)}$ is achieved to constrain feature structures at different levels

$$L_{depth} = \sum_{i=1}^{3} L_{dice}(R^{(i)}(PL_2), Fs^{(i)}). \qquad (6)$$

where $R^{(i)}(PL_2)$ represents the supervision label from layer i, with the same size as $Fs^{(i)}$.

D. Loss Function

The total loss of FDDSeg is composed of four parts: L_{pce}, L_{pls}, L_{depth} and L_{fdd}

$$L_{total} = \lambda_1 \cdot L_{pce} + \lambda_2 \cdot L_{pls} + \lambda_3 \cdot L_{depth} + \lambda_4 \cdot L_{fdd}. \qquad (7)$$

where λ_1, λ_2, λ_3, and λ_4 are employed to balance the weights of different loss.

IV. Experiments and Results

A. Dataset

The two public datasets ACDC [30] and MSCMR [10] are adopted to validate the effectiveness of FDDSeg. The ACDC dataset collected magnetic resonance imaging (MRI) scans (1902 slices) from 100 patients at the end-diastolic and end-systolic phases, each slice is affiliated with scribble and pixel annotation on left ventricles (LV), right ventricles (RV), and myocardium (MYO). Following [21], the ACDC dataset is objectively divided into training, validation, and testing sets with 7:1.5:1.5 ratio. The MSCMR dataset is composed of late gadolinium enhancement cardiac MRI scans (686 slices) obtained from 45 patients with cardiomyopathy, among which 20 sets of MRI images are provided scribble and pixel annotations of the LV, RV, and MYO simultaneously. The other 25 sets of MRI images are only provided with scribble annotations of the

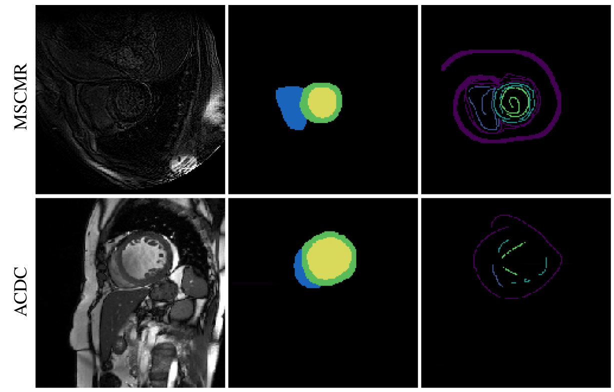
Algorithm 1: Algorithm of FDD.

Input: $Ft^{(i)}$, $Fs^{(i)}$, PL_2, the number of classes n, the hyper-parameters α, the index set of classes $\Phi_n = \{0, 1, \ldots, n-1\}$
Output: ${L_{fdd}}^{(i)}$

$l\_hot = onehot(PL_2)$
$Ft\_soft = softmax(Ft^{(i)})$
$Fs\_soft = softmax(Fs^{(i)})$
for each $ii \in range(n)$ **do**
  $ft_{ii} = Ft^{(i)}[:, ii, :, :]$, $fs_{ii} = Fs^{(i)}[:, ii, :, :]$
  $ft\_soft_{ii} = Ft\_soft[:, ii, :, :]$
  $fs\_soft_{ii} = Fs\_soft[:, ii, :, :]$
  $l_{ii} = l\_hot[:, ii, :, :]$
  $\bar{l_{ii}} = |l_{ii} - 1|$
end for
for each $j \in range(n)$ **do**
  $ft\_t_j = ft\_soft_j * l_j$, $fs\_t_j = fs\_soft_j * l_j$
  $ft\_nt_j = ft\_soft_j * \bar{l_j} + \sum_{k \in \Phi_n, k \neq j} ft\_soft_k$
  $fs\_nt_j = fs\_soft_j * \bar{l_j} + \sum_{k \in \Phi_n, k \neq j} fs\_soft_k$
  $\Phi = \Phi_n - \{j\}$
  $ft\_nct_j = [ft_j * \bar{l_j}, ft_{\Phi(0)}, \ldots, ft_{\Phi(n-2)}]$
  $fs\_nct_j = [fs_j * \bar{l_j}, fs_{\Phi(0)}, \ldots, fs_{\Phi(n-2)}]$
  $Ft\_t^{(i)} = Ft\_t^{(i)}.append(ft\_t_j)$
  $Fs\_t^{(i)} = Fs\_t^{(i)}.append(fs\_t_j)$
  $Ft\_nt^{(i)} = Ft\_nt^{(i)}.append(ft\_nt_j)$
  $Fs\_nt^{(i)} = Fs\_nt^{(i)}.append(fs\_nt_j)$
  $Ft\_nct^{(i)} = softmax(Ft\_nct^{(i)}.append(ft\_nct_j))$
  $Fs\_nct^{(i)} = softmax(Fs\_nct^{(i)}.append(fs\_nct_j))$
end for
$L_{class} = L_{mse}([Ft\_t^{(i)}, Ft\_nt^{(i)}], [Fs\_t^{(i)}, Fs\_nt^{(i)}])$
$L_{class\_free} = L_{mse}(Ft\_nct^{(i)}, Fs\_nct^{(i)})$
${L_{fdd}}^{(i)} = (1 - \alpha) * L_{class} + \alpha * L_{class\_free}$
return Output

LV, RV, and MYO. Following [10], the MSCMR dataset is objectively divided into training, validation, and testing with 5:1:3 ratio. Random rotation and flipping were applied to augment the training set, with the augmented image resized to 256×256. Fig. 4 demonstrates two exemplar images and their annotations from the two datasets.

The selection of labeled pixels follows the following principles. For simple samples, the segmentation challenge mainly focuses on the boundary parts of different classes to be segmented, the labeled pixels are mainly located at the boundaries of different segmentation classes. For complex samples, besides the boundary part, segmentation challenges also exist within the area to be segmented, which may be segmentation errors caused by color and texture changes. For them, the labeled pixels are mainly chosen near the boundaries of different classes to be segmented and within an appropriate number of classes to be segmented. In addition, we also achieve annotation enhancement by increasing the number or width of annotation lines.



(a) Images (b) Masks (c) Scribbles

Fig. 4. Examples from MSCMR and ACDC, include images, masks, and scribbles. Since the structural boundaries in MSCMR images are more difficult to distinguish, a thicker scribble is used to annotate the background for enhanced supervision.

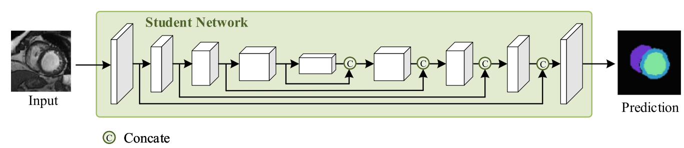


Fig. 5. The test workflow. The testing process involves solely the student network. Raw images are fed into the student network, where segmentation results are generated automatically without any additional steps.

B. Implementation Details

FDDSeg is implemented with Pytorch on Intel(R) Xeon(R) W-2175 CPU and 1 NVIDIA 2080Ti GPU. We apply the SGD optimizer (weight decay = 0.0004, momentum = 0.9) to train the model. The poly learning rate approach was employed to adjust the learning rate online [14]. The initial learning rate, batch size, and iteration are set to 1×10^{-2}, 4, and 6×10^4, respectively. We empirically set $\lambda_1 = 1$ and $\lambda_2 = 0.5$. The values of λ_3 and λ_4 are determined by searching within some empirically defined value sets/ranges. Specifically, they are searched within the value range [0, 0.5]. The final hyperparameters λ_1 to λ_4 are set to $\{1, 0.5, 0.35, 0.15\}$ and $\{1, 0.5, 0.05, 0.05\}$ in the ACDC and MSCMR datasets, respectively.

The test phase is shown in Fig. 5. We input the raw image into the student branch encoder for feature extraction, then perform up-sampling resolution restoration through the decoder, and finally obtain segmentation in the final output layer.

C. Evaluation Metrics

Dice Similarity Coefficient (DSC), Jaccard Index (JC), and 95% Hausdorff Distance (HD95) are referred to as evaluation metrics. DSC and JC can represent the overlap between labels and the volume of output results. HD95 is capable of indicating the boundary distance between the volume of prediction results and labels. For the prediction A and ground truth B, the DSC

TABLE I
COMPARISON WITH DIFFERENT FULLY-SUPERVISED METHODS

Methods	Type	MSCMR			ACDC		
		DSC	JC	HD95	DSC	JC	HD95
Unet_F	FSL	84.83	73.72	6.22	87.26	77.47	4.22
Unet_F^+	FSL	85.30	74.69	5.89	87.83	78.38	4.37
Unet_F^{++}	FSL	85.94	75.51	6.07	88.18	78.87	3.13
Puzzle Mix_F	FSL	86.90	76.96	4.25	88.51	79.39	3.80
TransUnet	FSL	81.24	68.75	4.06	89.89	81.81	2.38
SwinUnet	FSL	83.21	71.31	3.43	90.28	82.39	2.96
CSUnet	FSL	**89.14**	**81.48**	**1.71**	**90.91**	**83.45**	**1.03**
FDDSeg (Ours)	WSL	88.75	79.78	4.44	89.05	80.30	5.76

[b]Best results are indicated in Bold.

metric is formulated as follows:

$$DSC(A,B) = \frac{2\,|A \cap B|}{|A| + |B|} \times 100\%. \tag{8}$$

and the detail of JC metric is as follows:

$$JC(A,B) = \frac{|A \cap B|}{|A \cup B|} \times 100\%. \tag{9}$$

To obtain the HD95 metric, we should define the maximum hausdorff distance $d_H(A,B)$ at first. The $d_H(A,B)$ is the maximum distance of a set A to the nearest point in the other set B, defined as:

$$d_H(A,B) = \max\{\max_{a\in A}\min_{b\in B} d(a,b), \max_{b\in B}\min_{a\in A} d(a,b)\}. \tag{10}$$

HD95 is similar to $d_H(A,B)$, and it is based on the calculation of the 95% of the distance between boundary points in A and B.

D. Results

1) Comparison With Fully-Supervised Methods: Table I provides the comparison between FDDSeg and some fully-supervised methods on MSCMR and ACDC datasets. Unet_F[31], Unet_F^+[32], and Unet_F^{++}[15] are implemented based on convolutional neural networks, achieving acceptable segmentation by basic structure. Through data augmentation, Puzzle Mix_F[33] improves the DSC score of Unet_F^+ from 85.3% to 86.9% on MSCMR, and from 87.83% to 88.51% on ACDC. As the first work to utilize Transformer for medical image segmentation, TransUnet [17] maintains the similar U style of the Unet and employed Transformer basic layers followed by convnets for feature extraction, obtaining an obvious improvement compared with the three mentioned Unet methods on the ACDC dataset. Inspired by Swin-transformer, SwinUnet [2] keeps the similar U style of the Unet and implemented hierarchical feature extraction with shifted windows, drastically reducing the quadratic complexity of traditional self-attention mechanism and achieving better feature expression. Implementing multi-head attention in the transformer using convolution layers, the CS-Unet [34] is capable of capturing both local and global information and achieves best segmentation among the 8 methods. By contrast, FDDSeg could obtain the competitive performance, and the margins are so exciting because it did in the scribble supervision. This indicates that with a specific design, FDDSeg has great potential for application in actual medical scenarios at a lower annotation cost.

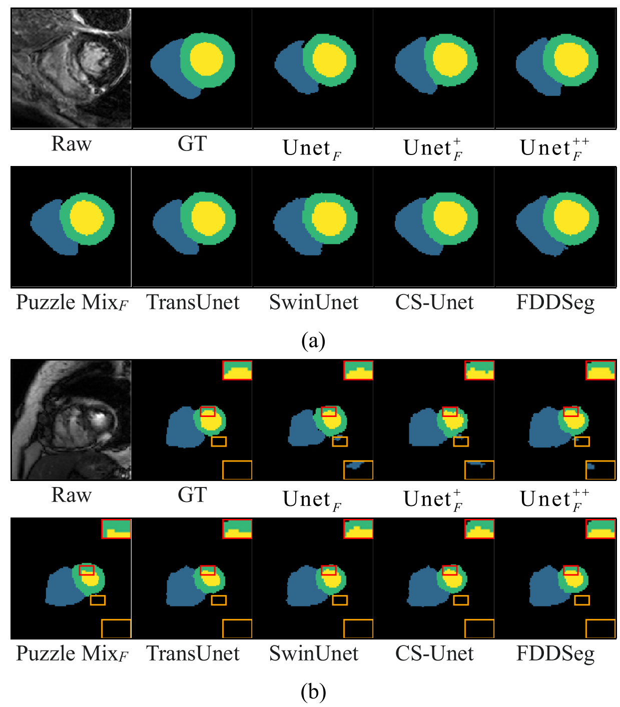


Fig. 6. Qualitative comparison between FDDSeg and other methods on MSCMR(a) and ACDC(b) datasets. During the training phase, the FDDSeg and other methods employ scribbles and masks as supervisory signals, respectively.

Fig. 6 provides segmentation visualizations between FDDSeg and some fully-supervised methods on MSCMR and ACDC datasets. Unet_F, Unet_F^+, and Unet_F^{++} achieve the basic segmentation due to the lack of global information, resulting in inaccurate boundary segmentation and false positives in segmentation. Puzzle Mix_F alleviates segmentation instability caused by insufficient sample number by ensuring that the mixed image contains sufficient target information for each mixed input while retaining local statistical information for each mixed input. TransUnet, a fusion of CNN and transformer, excels in capturing both local and global information concurrently, thereby reducing false positives in segmentation. SwinUnet, incorporating shift window operations, efficiently captures intricate features across

TABLE II
COMPARIHSON WITH STATE-OF-THE-ART SCRIBBLE-SUPERVISED SEGMENTATION METHODS ON THE MSCMR AND ACDC DATASET

Methods	MSCMR			ACDC		
	DSC	JC	HD95	DSC	JC	HD95
MAAG	84.51	73.18	7.8	83.40	71.66	9.5
ShapePU	85.20	74.41	15.0	83.55	71.83	12.6
	(0.69 ↑)	(1.23 ↑)	(7.2 ↓)	(0.15 ↑)	(0.17 ↑)	(3.1 ↓)
CycleMix	80.00	66.86	19.1	84.87	73.77	17.9
	(4.51 ↓)	(6.32 ↓)	(11.3 ↓)	(1.47 ↑)	(2.11 ↑)	(8.4 ↓)
DB	88.45	79.33	8.10	88.33	79.13	11.15
	(3.94 ↑)	(6.15 ↑)	(0.3 ↓)	(4.93 ↑)	(7.47 ↑)	(1.65 ↓)
ZS	87.00	77.09	26.5	86.20	75.79	9.79
	(2.49 ↑)	(3.91 ↑)	(18.7 ↓)	(2.8 ↑)	(4.13 ↑)	(0.29 ↓)
PDB	88.64	79.71	5.67	89.02	80.27	8.51
	(4.13 ↑)	(6.53 ↑)	(2.13 ↓)	(5.62 ↑)	(8.61 ↑)	(0.99 ↓)
ScribbleVC	86.80	76.76	10.03	88.40	79.24	8.84
	(2.29 ↑)	(3.58 ↑)	(2.23 ↓)	(5.00 ↑)	(7.58 ↑)	(0.66 ↓)
FDDSeg (ours)	**88.75**	**79.78**	**4.44**	**89.05**	**80.30**	**5.76**
	(4.24 ↑)	**(6.60 ↑)**	**(3.36 ↑)**	**(5.65 ↑)**	**(8.64 ↑)**	**(3.74 ↑)**

[c]Best results are indicated in Bold. The numbers in parentheses represent the gain of the current metric compared to the corresponding metric of MAAG, where ↑ is a positive gain and ↓ is a negative gain.

diverse image regions. CS-Unet, a convolutional variant of SwinUnet, minimizes reliance on linear layers while preserving detailed features and retaining feature space information, leading to superior segmentation outcomes. FDDSeg employs regional-level supervision via a mixed prediction framework of distinct pseudo labels. It leverages convolution to capture local details and compensates for global information limitations by integrating diverse branch perspectives, thereby mitigating redundant segmentation. Additionally, through deep supervision technique, FDDSeg systematically shapes feature maps layer by layer, alleviating boundary noise.

2) Comparison With Scribble-Supervised Methods: We compared our FDDSeg with 7 state-of-the-art methods: MAAG [21], ShapePU [11], CycleMix [10], Dual-Branch (DB) [14], ZscribbleSeg (ZS) [12], Perturbation Dual-Branch (PDB) [35], and ScribbleVC [36]. Table II displays the comparison results on ACDC and MSCMR testing sets. It can be seen that FDDSeg achieved the ideal score on DSC, JC, and HD95. MAAG introduces the generative adversarial method, and discriminates the segmentation results by the unpaired real label, leading to the unsatisfied segmentation accuracy. ShapePU and CycleMix apply random occlusion to process training labels, which makes it easy to lose limited annotation information and weaken scribble supervision. Ignoring knowledge learning between different branches, the Dual-Branch and Perturbation Dual-Branch are hard to achieve optimal segmentation. ZScribbleSeg mainly focuses on issues such as network supervision and prediction correction, lacking further exploration of network structure. ScribbleVC has designed a hybrid encoder based on CNN and Transformer. However, the Transformer is prone to insufficient training on small-scale datasets, resulting in insufficient resolution of extracted features and limited overall model feature expression. Our FDDSeg benefits from FDD, enabling feature distillation between different branches, and owns reliable supervision with PLMS, effectively compensating for the limitations of scribble supervision at the regional and boundary levels, thus it achieves super segmentation.

Fig. 7 also visualize the segmentation results of different methods on MSCMR as well as ACDC. We can observe that MAAG construes scribble segmentation by unpaired real labels, which provides limited identification information, leading to unsatisfactory boundary segmentation. ShapePU and CycleMix, due to the introduction of random occlusion, are prone to damaging the original image structural information, resulting in under-segmentation. Dual-Branch and Perturbation Dual-Branch suffer from weak discrimination in difficult-to-segment area since they ignore the information interaction between different branches. ScribbleVC adopts an encoder that takes the mixed feature from CNN and Transformer as feature input, which easily makes the model hard to distinguish key feature regions, resulting in a few redundant segmentation and shape defects on small-scale dataset MSCMR. FDDSeg utilizes SR to obtain accurate boundary supervision, alleviating redundant segmentation on the MSCMR dataset, and maintaining the shape structure of different segmentation regions on the ACDC dataset, which makes it outperform other competing methods. In addition, L_{depth} improves the accuracy of boundary segmentation by constraining the shape of feature maps layer by layer.

To further analyze the performance of the model, we calculated the parameters and average inference time of compared methods, as shown in Table III. It can be concluded that our

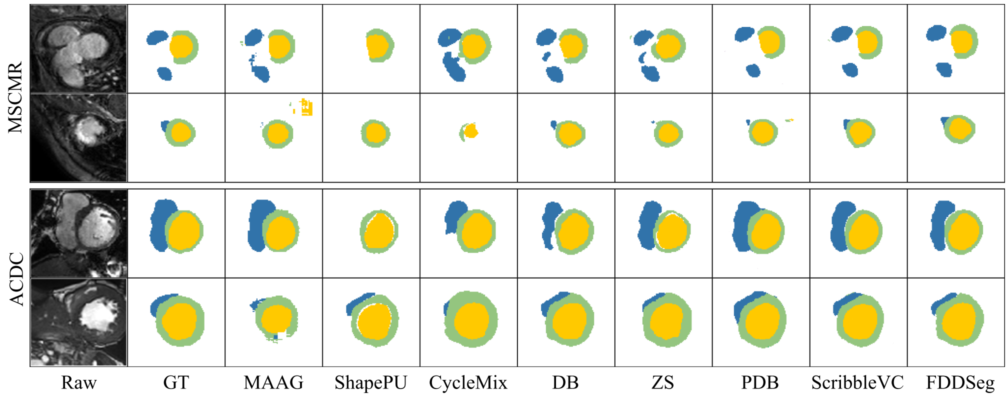


Fig. 7. Qualitative comparison between FDDSeg and other scribble-supervised methods on MSCMR and ACDC datasets. During the training phase, all method uses scribbles as supervisory signals. Each dataset provides segmentation results of two images for display.

TABLE III
COMPARISON WITH SEVERAL MODELS ON PARAMETERS AND INFERENCE TIME

Methods	Hyper-parameters (M)	Inference-time (s)
MAAG	21.74	0.139
ShapePU	25.17	0.202
CycleMix	25.17	0.175
DB	2.720	0.069
ZS	25.03	0.166
PDB	3.117	0.076
ScribbleVC	47.80	0.346
FDDSeg (ours)	**2.010**	**0.048**

[d]Best results are indicated in Bold.

method relies on the fewest parameters, the smallest storage, but the fastest inference. This mainly thanks an independent self-teacher and student model that can be saved separately. At the same time, the design of decomposition distillation is capable of extracting relatively complete features with fewer channels. In the comparison model, MAAG, ShapePU, CycleMix, and ZscribbleSeg are implemented based on the UNet, and a large number of feature channels are often needed for better performance, but it produces heavier computation load. Dual Branch devised a structure featuring a single encoder and dual decoder. Its testing process requires the participation of all parameters of the single encoder and dual decoder, thereby the model parameters can not be reduced. ScribbleVC has developed a hybrid design based on CNN and a transformer, and the structural of the Transformer itself makes it difficult to reduce the parameters.

V. DISCUSSION

A. The Determination of α

For optimization, parameter α was set in [0, 1] with a stepsize of 0.1, and the resulting DSCs on both ACDC and MSCMR are shown in Fig. 8. It can be observed that the DSC curve first increases and then decreases with the increase of α. The peak was reached with α higher than 0.5, indicating that the distillation gain can be further improved by assigning proper weights to the class and class-free features. According to the experimental results, to appropriately reduce optimization work, we select the

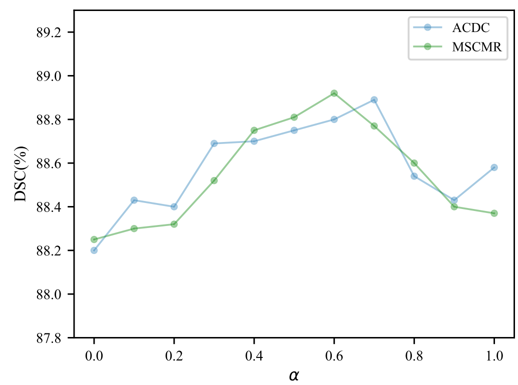


Fig. 8. Change of DSC scores under different values of α on MSCMR and ACDC datasets.

TABLE IV
ABLATION STUDY OF FDD DECOMPOSITION LABELS

Strategy	MSCMR			ACDC		
	DSC	JC	HD95	DSC	JC	HD95
$\hat{y}_1$	88.74	79.76	6.89	88.46	79.33	9.42
PL_2	**88.75**	**79.78**	**4.49**	**89.05**	**80.30**	**5.76**

[b]Best results are indicated in Bold.

α values corresponding to the highest DSC score and the second highest DSC score, respectively, and calculate their mean as the final α, which means that α is set to 0.6 and 0.55 for ACDC and MSCMR respectively.

B. The Determination of FDD Composition Label

To verify the rationality of PL_2, we conducted experiments using PL_2 and $\hat{y}_1$ as decomposition labels for FDD, with the results outlined in the Table IV. As depicted in the Table IV, compared to PL_2, $\hat{y}_1$ weakens segmentation performance. This is because using $\hat{y}_1$ as a decomposition label is equivalent to having the teacher branch guide the common part (P) and the student feature selected by the teacher's individual part (m_2 and m_4), as shown in Fig. 9(a). Our objective is to devise a self-distillation model wherein both the teacher and student models adopt similar network structures. The teacher model's

TABLE V
COMPARISON WITH DIFFERENT EXPERIMENTAL STRATEGIES

Methods	Loss					MSCMR			ACDC		
	L_{pce}	L_{pls}	L_{depth}	L_{KD}	L_{fdd}	DSC	JC	HD95	DSC	JC	HD95
Backbone	√	√	√			88.46	79.41	5.50	88.68	79.86	8.33
Backbone+KD	√	√	√	√		88.54	79.44	5.07	88.82	79.89	7.36
FDDSeg (w/o PLMS)	√		√		√	52.35	36.46	191.7	54.35	37.33	197.2
FDDSeg (w/o SR)		√	√		√	2.530	1.370	256.3	2.820	1.610	123.5
FDDSeg (w/o PL_1)	√	√	√		√	88.50	79.43	4.90	**89.15**	**80.44**	9.53
FDDSeg	√	√	√		√	**88.75**	**79.78**	**4.44**	89.05	80.30	**5.76**

[a]Best results are indicated in Bold.

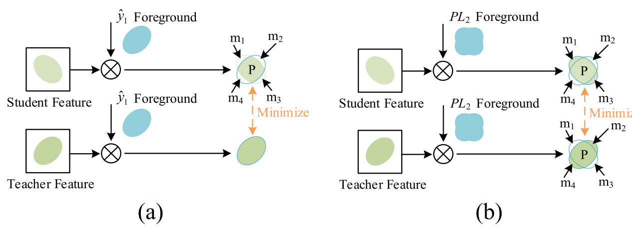


Fig. 9. Decomposition results with different decomposition labels. The symbol ⊗ represents the element-wise multiplication. We employ the $\hat{y_1}$ (a) and PL_2 (b) as decomposition labels, respectively, to generate decomposition results for the features of teachers and students.

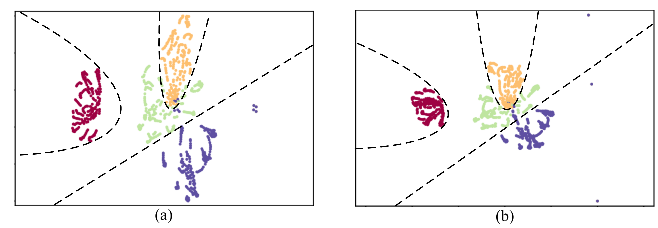


Fig. 10. t-SNE of feature learned for KD(a) and FDD(b). The relationship between segmentation classes and colors: Background (red), RV (orange), MYO (green), and LV (blue). The dashed line represents the segmentation boundary between different classes.

role is to offer students an alternative segmentation perspective, mitigating overconfident predictions. The feasibility of PL_2 lies in that it can reduce the confidence prediction of individual parts of student (m_1 and m_3) and enhance their learning of individual parts of teacher (m_2 and m_4), as shown in Fig. 9(b). This design encourages two branches to optimize towards the same segmentation goal, thus achieving overall self-learning of the model and improving segmentation ability.

C. Ablation Study

1) The Effectiveness of Different Module: In our experiment, FDDSeg without an FDD module is taken as the baseline. We explored the effectiveness of classical knowledge distillation (KD) [22], PLMS, SR, and FDD respectively, and the results are shown in Table V.

It is shown that the classical KD only brings limited performance improvement, which may be because KD usually acts on the later stage of model training. However, it is shown that the model is more likely to capture the correct classification knowledge in the early stage of training, and the knowledge is easily forgotten in the later stage of training, resulting in the neglect of early knowledge and difficulty in obtaining maximum gain [37]. SR is the only module that can provide limited but accurate label for model training. It can help locate the main position of the segmented area. Without SR, the model can only perform self-supervision through prediction under irregular learning, making it difficult to train effectively, which may deteriorate the segmentation. Considering the annotations provided by SR can only supervise a few pixels, which makes it difficult to achieve supervision at the regional and boundary levels. PLMS is further designed. It dynamically integrates the predictions of teacher branch and student branch, as well as pseudo labels generated based on superpixel algorithms and scribble information, which helps achieve better supervision at the regional and boundary layer levels. Both SR and PLMS focus on mining existing supervisory information to provide more guidance at the boundary and regional levels for the model, which can significantly improve the segmentation performance of FDDSeg. Thus, the significant decrease will show up if SR or PLMS is excluded from the model. With the decomposition of the intermediate features, the FDD constrained the shape of the target by the class features, and captured the correlation between classes by the class-free features, effectively enhancing the features. To further verify the effectiveness of FDD, t-SNE analysis was applied for the intermediate features in the MSCMR test set. From Fig. 10, it is easy to see that FDD achieves better aggregation, and the segmentation challenge mainly lies between the classes represented by yellow pixels and green pixels. It can be observed that, in comparison to KD, the segmentation boundary of the class represented by yellow pixels, as fitted by FDD, contains fewer green pixels. This implies that FDD effectively diminishes the similarity between classes represented by RV and MYO, thereby mitigating segmentation challenges.

2) The Effectiveness of Different Loss Function: To further validate the effectiveness of the proposed method, we have conducted ablation experiments on different loss functions. The relevant experimental results are shown in the Table VI. L_{pce} provides limited yet precise supervision via scribble information, enabling the model to achieve a basic segmentation result. PLMS provides rich region and boundary supervision information for the model by dynamically integrating three types of pseudo labels (PL_1, $\hat{y_1}$, $\hat{y_2}$). Therefore, it decreases the HD95 of the proposed model from 17.09 to 5.50 on MSCMR, and from 8.68 to 8.33 on ACDC. L_{depth} further alleviates the phenomenon of boundary outliers by constraining intermediate features layer by layer. L_{fdd} further captures inter-class correlations through more attention to class-free features, facilitating distillation

TABLE VI
ABLATION STUDY OF LOSS FUNCTIONS

L_{pce}	L_{pls}	L_{depth}	L_{fdd}	MSCMR			ACDC		
				DSC	JC	HD95	DSC	JC	HD95
√				58.90	41.82	159.74	43.32	27.87	177.62
√	√			87.96	78.71	17.09	88.42	79.30	8.68
√	√	√		88.46	79.41	5.50	88.68	79.86	8.33
√	√	√	√	**88.75**	**79.78**	**4.44**	**89.05**	**80.30**	**5.76**

[b]Best results are indicated in Bold.

learning, thus enabling the proposed method to achieve optimal segmentation.

VI. Conclusion

This paper implements the feature decomposition distillation method (FDDSeg) for segmenting cardiac MRI images utilizing scribbling annotation. Using the pseudo-labels from the model, the hidden layer features are broken down into class features and class-free features. Then, feature learning is improved following the fundamental learning principle of class-free features, which fully utilizes distillation knowledge based on the teacher-student distillation framework. To improve the supervisory signal, a clustering method is also implemented to generate pseudo-labels from unlabeled pixels. When compared to the 7 sophisticated scribble-supervised approaches, experimental results on the ACDC and MSCMR datasets show that FDDSeg achieves comparable segmentation performance with the fewest model parameters, which makes it easier to deploy in medical instruments for further real-world application.

Despite FDDSeg's notable improvement in segmentation performance owing to its lightweight architecture, it falls short in providing comprehensive intermediate feature representations. Our aim is to bolster FDD modules' efficacy by refining network structures to enhance these intermediate feature representations. Enhanced feature representations promise to provide richer distillation knowledge, thus proving advantageous for FDD. To mitigate the rise in model parameters, future work could explore combining feature swap with perturbation techniques like noise perturbation and dropout. This synergistic approach holds promise for further enhancing FDDSeg's performance while preserving its lightweight nature.

Nevertheless, our method still suffers from the annotation workload, requiring annotation for every MRI picture used in training. To achieve optimal segmentation with minimal annotation, future research will investigate the integration of self-supervised learning with scribbling segmentation.

References

[1] H. Huang et al., "UNet 3+: A full-scale connected unet for medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2020, pp. 1055–1059.

[2] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. 17th Eur. Conf. Comput. Vis.*, 2022, pp. 205–218.

[3] Y. Gu, Y. Guan, Z. Yu, and B. Dong, "SegCoFusion: An integrative multimodal volumetric segmentation cooperating with fusion pipeline to enhance lesion awareness," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 12, pp. 5860–5871, Dec. 2023.

[4] Q. Hou, Y. Peng, Z. Wang, J. Wang, and J. Jiang, "MFD-Net: Modality fusion diffractive network for segmentation of multimodal brain tumor image," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 12, pp. 5958–5969, Dec. 2023.

[5] Y. Zhao, X. Shao, C. Chen, J. Song, C. Tian, and W. Li, "The contrastive network with convolution and self-attention mechanisms for unsupervised cell segmentation," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 12, pp. 5837–5847, Dec. 2023.

[6] H. Chen, J. An, B. Jiang, L. Xia, Y. Bai, and Z. Gao, "WS-MTST: Weakly supervised multi-label brain tumor segmentation with transformers," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 12, pp. 5914–5925, Dec. 2023.

[7] S. Zhai, G. Wang, X. Luo, Q. Yue, K. Li, and S. Zhang, "PA-Seg: Learning from point annotations for 3D medical image segmentation using contextual regularization and cross knowledge distillation," *IEEE Trans. Med. Imag.*, vol. 42, no. 8, pp. 2235–2246, Aug. 2023.

[8] Q. Chen and Y. Hong, "Scribble2D5: Weakly-supervised volumetric image segmentation via scribble annotations," in *Proc. 25th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2022, pp. 234–243.

[9] W. Li, W. Liu, J. Zhu, M. Cui, X.-S. Hua, and L. Zhang, "Box-supervised instance segmentation with level set evolution," in *Proc. 17th Eur. Conf. Comput. Vis.*, 2022, pp. 1–18.

[10] K. Zhang and X. Zhuang, "CycleMix: A holistic strategy for medical image segmentation from scribble supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11646–11655.

[11] K. Zhang and X. Zhuang, "ShapePU: A new PU learning framework regularized by global consistency for scribble supervised cardiac segmentation," in *Proc. 25th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2022, pp. 162–172.

[12] K. Zhang and X. Zhuang, "ZscribbleSEG: Zen and the art of scribble supervised medical image segmentation," 2023, *arXiv:2301.04882*.

[13] Y. Can, K. Chaitanya, B. Mustafa, L. Koch, E. Konukoglu, and C. Baumgartner, "Learning to segment medical images with scribble-supervision alone," in *Proc. Deep Learn. Med. Image Anal. Multimodal Learn. Clin. Decis. Support*, 2018, pp. 236–224.

[14] X. Luo et al., "Scribble-supervised medical image segmentation via dual-branch network and dynamically mixed pseudo labels supervision," in *Proc. 25th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2022, pp. 528–538.

[15] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++ : A nested U-Net architecture for medical image segmentation," in *Proc. Deep Learn. Med. Image Anal. Multimodal Learn. Clin. Decis. Support*, 2018, pp. 3–11.

[16] J. Schlemper et al., "Attention gated networks: Learning to leverage salient regions in medical images," *Med. Image Anal.*, vol. 53, pp. 197–207, 2019.

[17] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[18] K. Wu, B. Du, M. Luo, H. Wen, Y. Shen, and J. Feng, "Weakly supervised brain lesion segmentation via attentional representation learning," in *Proc. 22th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2019, pp. 211–219.

[19] I. Laradji et al., "A weakly supervised consistency-based learning method for COVID-19 segmentation in CT images," in *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2021, pp. 2453–2462.

[20] Z. Zhao, L. Yang, H. Zheng, I. H. Guldner, S. Zhang, and D. Z. Chen, "Deep learning based instance segmentation in 3D biomedical images using weak annotation," in *Proc. 21th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2018, pp. 352–360.

[21] G. Valvano, A. Leo, and S. A. Tsaftaris, "Learning to segment from scribbles using multi-scale adversarial attention gates," *IEEE Trans. Med. Imag.*, vol. 40, no. 8, pp. 1990–2001, Aug. 2021.

[22] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” 2015, *arXiv:1503.02531*.

[23] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio, “FitNets: Hints for thin deep nets,” in *Proc. 3rd Int. Conf. Learn. Representations*, 2015, pp. 1–13.

[24] B. Zhao, Q. Cui, R. Song, Y. Qiu, and J. Liang, “Decoupled knowledge distillation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11943–11952.

[25] L. Zhang, J. Song, A. Gao, J. Chen, C. Bao, and K. Ma, “Be your own teacher: Improve the performance of convolutional neural networks via self distillation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 3712–3721.

[26] M. Ji, S. Shin, S. Hwang, G. Park, and I.-C. Moon, “Refine myself by teaching myself: Feature refinement via self-knowledge distillation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 10659–10688.

[27] J. Cheng et al., “DDU-Net: A dual dense U-structure network for medical image segmentation,” *Appl. Soft Comput.*, vol. 126, 2022, Art. no. 109297.

[28] P. F. Felzenszwalb and D. P. Huttenlocher, “Efficient graph-based image segmentation,” *Int. J. Comput. Vis.*, vol. 59, no. 2, pp. 167–181, 2004.

[29] H. Lee and W.-K. Jeong, “Scribble2label: Scribble-supervised cell segmentation via self-generating pseudo-labels with consistency,” in *Proc. 23th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2020, pp. 14–23.

[30] O. Bernard et al., “Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved?,” *IEEE Trans. Med. Imag.*, vol. 37, no. 11, pp. 2514–2525, Nov. 2018.

[31] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. 25th Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2015, pp. 234–241.

[32] C. F. Baumgartner, L. M. Koch, M. Pollefeys, and E. Konukoglu, “An exploration of 2D and 3D deep learning techniques for cardiac MR image segmentation,” in *Proc. Int. Workshop Stat. Atlases Comput. Models Heart*, 2017, pp. 111–119.

[33] J.-H. Kim, W. Choo, and H. O. Song, “Puzzle mix: Exploiting saliency and local statistics for optimal mixup,” in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 5275–5285.

[34] Q. Liu, C. Kaul, J. Wang, C. Anagnostopoulos, R. Murray-Smith, and F. Deligianni, “Optimizing vision transformers for medical image segmentation,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2023, pp. 1–5.

[35] J. Ying, W. Huang, L. Fu, H. Yang, and J. Cheng, “Weakly supervised segmentation of uterus by scribble labeling on endometrial cancer MR images,” *Comput. Biol. Med.*, vol. 167, 2023, Art. no. 107582.

[36] Z. Li, Y. Zheng, X. Luo, D. Shan, and Q. Hong, “ScribbleVC: Scribble-supervised medical image segmentation with vision-class embedding,” in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 3384–3393.

[37] S. Liu, K. Liu, W. Zhu, Y. Shen, and C. Fernandez-Granda, “Adaptive early-learning correction for segmentation from noisy annotations,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2596–2606.